

Meshtastic 915 MHz Yagi Antenna Build Instructions

Thank you for participating in the GCARC 915 MHz Yagi antenna group build program. While this is an extremely simple build, it could be confusing to anyone who does not fully understand the process. To help avoid any such confusion, we have put together this set of build instructions. Follow the step-by-step procedure below to achieve a proper build of the antenna kit.

1. Verify that all of the required parts are on hand, as listed below:
 - a. antenna boom;
 - b. antenna handle;
 - c. antenna window bracket with an acorn nut installed;
 - d. suction cup with a mounting stud attached;
 - e. set of pre-cut brass rod elements (six pieces, packaged inside plastic tube);
 - f. RG-174 coaxial cable 18" long with male SMA connector on one end;
 - g. two (2) small (4") zip ties;
 - h. four-inch length of 22AWG solid wire (various colors);
 - i. SMA thumb wrench;
 - j. 10-32 nylon fillister head machine screw;
 - k. 10-32 nylon wing nut;
 - l. two (2) #10 stainless steel flat washers; and
 - m. two (2) #10 externally-toothed lock washers.
2. Verify that the required supplies and equipment are on hand, as listed below:
 - a. soldering station with rosin-core solder;
 - b. 99.9% isopropyl alcohol (IPA);
 - c. paper towel(s);
 - d. cyanoacrylate (CA) glue;
 - e. accelerator for CA glue;
 - f. razor knife;
 - g. Sharpie® marker;
 - h. ruler or measuring tape;
 - i. push-pin or pointed tweezer;
 - j. needle-nose pliers;
 - k. wire cutter and stripper;
 - l. NanoVNA; and
 - m. 915 MHz Meshtastic Node.
3. If not already done, remove the brass rod elements from the packing tube, and using a paper towel and some IPA, thoroughly clean each element rod of all film and residue that may be on the element rods.
4. Lay out the element rods in order of decreasing length, setting the two shortest (equal length) element rods aside.
5. Measure and mark the center of each of the long element rods.

6. Make another pair of marks on each element rod, making one exactly $7/16$ " on each side of the center mark. This will define an area $7/8$ " wide at the center of each element rod.
7. Examine the boom carefully, noting the markings adjacent to each of the small ($1/16$ ") element holes in the boom. These markings are as follows:
 - a. REF – the reflector element;
 - b. DE – the driven element (this is where the two shortest but equal-length element rods will eventually be installed);
 - c. D1 – the first director element;
 - d. D2 – the second director element; and
 - e. D3 – the third director element.
8. Starting with the longest element rod at the REF hole **and skipping the DE hole,** working from the longest to the shortest of the long element rods, insert a rod into the REF hole (the longest element rod) and each of the D# holes (progressively shorter element rods), centering the boom between the paired marks on the element rods.
9. Verify that the element rods get shorter as you progress along the boom, starting at the REF end of the boom, **and also that the DE hole is empty.**
10. Working quickly, one by one slide each element out of its hole in the boom just until the center mark is visible, and apply a drop of CA glue to the element rod, and then immediately center the boom again **accurately** between the paired marks on the element rod, twisting the element rod as it is inserted.
11. After all four of the long element rods are glued and centered **accurately**, apply a small spray of CA glue accelerant to each of the element rods at the boom. This will cause the CA glue to completely set almost immediately.
12. Set the boom and elements aside for later use.
13. Locate the four-inch length of 22AWG solid wire from the parts kit.
14. Using the wire stripper, remove $1/2$ " of insulation from each end of the wire. **SAVE ONE OF THE PIECES OF INSULATION THAT YOU REMOVE – IT WILL BE USED LATER!**
15. Using the small end of the barrel of the Sharpie® marker as a spindle, form a single-turn loop in the center of the insulated portion of the wire, making a one-turn air-core inductor. The loop should be $1/4$ " in diameter with the leads extending straight out and opposite each other (in a straight line).
16. Apply a drop of CA glue to the inductor wire at the point where the wire crosses itself as it forms the loop.
17. Apply a bit of the CA glue accelerant to the inductor to quick-set the glue there.
18. Set the inductor aside for later use.
19. Locate the RG-174 coaxial cable.
20. Using the razor knife and working **carefully** so as not to cut the shield wire strands, remove $2-1/2$ " of outer jacket from the **unfinished** end of the cable.
21. Using the push pin, gently unbraid and comb out the shield wire strands, leaving them all parallel to the center conductor.

22. Once the shield wire strands are all separated and straightened, gather them all together and twist them tightly into a single bundle.
23. Using the wire stripper, carefully remove 1/2" of insulation from the coaxial cable center conductor. Work very carefully so as not to damage the center conductor wire strands.
24. Twist the center conductor wire strands together tightly.
25. Using the solder station, tin the twisted center conductor wire.
26. Using the solder station, tin the last 1/2" of the twisted shield wire.
27. Using the needle-nose pliers, bend a hook the size of the element rod into the end of each tinned segment of the coaxial cable wires.
28. Using the solder station, carefully solder the coaxial cable wire segments to the two short driven-element rods, attaching each of them about 1/2" from one end of each of the element rods, and allow the element rods to cool completely.
29. Locate the inductor that was set aside earlier.
30. Using the needle-nose pliers, bend a hook the size of the element rod into each stripped end of the inductor wire.
31. Using the solder station, carefully solder one end of the inductor wire to one of the driven-element rods, placing the inductor wire just outboard (towards the long end) of the element rod, and allow the element rod to cool completely.
32. Locate the piece of wire insulation (from the inductor wire) that you set aside earlier.
33. Cut off a small piece of this insulation, about 1/8" long.
34. Insert the 1/8" piece of insulation into the DE hole in the boom. Use the end of the push pin to push the insulation into the hole in the boom, pushing it about half-way through the boom.
35. Passing the looped inductor wire through the opening in the boom nearest the DE hole, insert one of the driven-element rods into the DE hole on each side of the boom and then position them so that each element rod is the same distance into the boom, and so that they meet in the middle **with the insulation piece in between the ends of the element rods.**
36. Solder the loose end of the inductor wire to the adjacent element rod immediately outboard of the coaxial cable wire attachment there, taking care not to let the coaxial cable wire separate from the element rod as you solder the inductor wire.
37. One at a time, slide the driven element rods out of the boom and apply a drop of CA glue to the end of the rod. Immediately insert the element rod back into its hole in the boom **without disturbing the opposite element rod.**
38. After both driven element rods have been glued, apply the CA glue accelerator spray to the boom and driven element rods to quick-set the glue.
39. Dress the coaxial cable along the boom adjacent to the DE hole in the boom, and secure it there with both of the 4" zip ties provided. **DO NOT RUN THE FULL LENGTH OF THE CABLE ALONG THE BOOM – LET IT HANG FREELY AFTER SECURING IT TO THE BOOM NEAR ITS ATTACHMENT TO THE DRIVEN ELEMENT RODS.** The zip ties should pass through the opening in the boom, and

should be pushed all the way towards and against the boom opening wall adjacent to the DE hole. The purpose of the zip ties is to support the coaxial cable and to take its weight off of the solder joints.

40. Locate the suction cup and the window bracket.
41. Thread the mounting stud on the suction cup into the acorn nut that is installed into the window bracket.
42. Locate the nylon machine screw, the nylon wing nut, the two #10 flat washers, and the two #10 externally-toothed lock washers.
43. Place one of the externally-toothed lock washers onto the nylon machine screw, and then place one of the flat washers onto the screw. The lock washer should be directly against the head of the screw.
44. Insert the boom into the yoke of the window bracket, being sure to position the boom so that the element rods will be parallel to the axis of the nylon machine screw once the screw is installed.
45. Insert the nylon machine screw through the ears of the window bracket yoke and also through the hole in the end of the boom, which will be aligned with the yoke ear holes.
46. Place the second flat washer onto the nylon machine screw and against the second ear of the window bracket yoke.
47. Place the second lock washer onto the nylon machine screw and against the second flat washer just installed.
48. Thread the nylon wing nut onto the nylon machine screw, taking care not to cross-thread the wing nut.
49. Tighten the nylon wing nut sufficiently to clamp the boom in the window bracket yoke. **DO NOT OVERTIGHTEN, STRIP, STRETCH, OR BREAK THE NYLON WING NUT AND/OR MACHINE SCREW.**

This completes the assembly of the antenna and its window bracket. Should you choose to use the antenna on the handle instead of on the window bracket, simply install the handle in place of the window bracket, being sure to install the hardware in the same manner. The flat washers must go against the yoke ears of either the handle or the window bracket, whichever is in use.

The antenna can now be tested using your NanoVNA and also by connection of the antenna to the 915 MHz Meshtastic node and placing the node into operation.